

JOHN MARK GATCHE

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SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, C++, C#, Java, Kotlin, LaTeX, PHP, HTML, CSS
Libraries & Frameworks:	React.js, Next.js, Tailwind CSS, FastAPI, Jetpack Compose
Data Science & ML:	Jupyter Notebook, RStudio, scikit-learn, pandas, NumPy
Design & Visualization:	Adobe Photoshop, Adobe Illustrator, Blender, Figma, Microsoft PowerPoint
Tools & Platforms:	Git, GitHub, IntelliJ IDEA, PyCharm, WebStorm, Visual Studio Code

EDUCATION

University of the East – Manila <i>BS Computer Science</i>	Graduated June 2026
▪ DOST-SEI Scholar and University Scholar GWA: 1.08 (Summa Cum Laude)	

EXPERIENCE

Big Pond Education <i>Frontend Developer Intern</i>	June 2025 – August 2025
▪ Developed responsive interfaces for a prototype barangay management system in Brgy. Sikatuna Village, Quezon City with TypeScript , React.js , and Tailwind CSS and resolved database/API bugs in the Laravel and SQLite backend.	
▪ Presented the prototype to four barangay offices (Sikatuna Village, San Miguel, West Triangle, Botocan), generating interest from local administrators as a potential solution for streamlining resident request processing.	
▪ Continued post-internship work migrating the database to PostgreSQL .	

LEADERSHIP & ACTIVITIES

Junior Philippine Computer Society – UE Manila Chapter <i>Dept. Head of Software Development</i>	July 2025 – April 2026
▪ Led development of the organization's official website (jpcs-ue.tech).	
▪ Collaborated with student officers to implement website updates supporting organizational initiatives.	
UE CCSS Research & Development Unit <i>Research Committee Assistant</i>	August 2023 – June 2024
▪ Coordinated and facilitated online cybersecurity webinars to enhance awareness among students.	
▪ Assisted in planning and executing college week competitions promoting math and computer science.	

PROJECTS

LakadPH – Pedestrian-First Smart Navigation Web App <i>Frontend Developer</i>
▪ Built a responsive frontend prototype with TypeScript , React.js , and Tailwind CSS , integrating OpenStreetMap for map data, ShadeMap API for shade coverage, and Overpass API for streetlight location data.
▪ Won Champion (1st Place) at Blue Hacks 2026 Hackathon held at Ateneo de Manila University.
Scaffl.ed – LLM-Driven Java Debugging Tutor via Gaze & Mouse Tracking <i>Project Team Leader</i>
▪ Led a five-member team in developing an intelligent tutoring system using TypeScript , React.js , Next.js , PostgreSQL , and FastAPI , managing project planning, research writing, and technical execution.
▪ Spearheaded the collection and curation of gaze/mouse interaction datasets and developed a Random Forest confusion detection model in Python using scikit-learn .
▪ Co-authored and presented the research at the 2026 International Conference on Advances in AI and Machine Learning (AAIML) in Tokyo, Japan, receiving the Outstanding Oral Presentation Award; publication indexed in IEEE Xplore and Scopus (DOI: 10.1109/AAIML.67890.2026.11498122).
Studdy – Study Partner Matching Application <i>Lead Android Developer</i>
▪ Led development of an Android study partner matching application, architecting the data layer with a repository pattern and dependency injection, and building the UI with Jetpack Compose under an MVVM architecture.
▪ Built custom UI components using Material 3 and low-level Compose drawing APIs to achieve specific design requirements.
▪ Integrated the app with a pre-built ASP.NET backend via Retrofit , handling API requests, response parsing, and error handling to ensure reliable data sync with the UI.
▪ Achieved 2nd Runner-Up, 2024 Philippine Startup Challenge – NCR Regional Pitching Competition.

ADDITIONAL CERTIFICATIONS & AWARDS

- IT Specialist Certification (Python) – Certiport, 2024
- 1st Runner-Up – Java Programming, University of Makati 12th IT Skills Olympics, 2023
- 1st Runner-Up – IT Quiz Bee, University of Makati 13th IT Skills Olympics, 2024
- Top 10 Finalist – Byte Forward Hackathon, 2025
- Champion – Public Administration Category, HABI Innovation Workshop, 2025
- Champion – IT Quiz Bee, University of Makati 14th IT Skills Olympics, 2025